```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

import warnings
warnings.filterwarnings('ignore')
```

```
!gdown 1BiZPRpiOCS9mpESuHB8SK7r8SXZoClod
```

```
Downloading...
From: https://drive.google.com/uc?id=1BiZPRpiOCS9mpESuHB8SK7r8SXZoClod
To: /content/ABS Tracking _ FY27 Offline - Partner_RAW (1).csv
100% 2.63M/2.63M [00:00<00:00, 137MB/s]
```

```
df = pd.read_csv("ABS Tracking _ FY27 Offline - Partner_RAW (1).csv")
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 14544 entries, 0 to 14543
Data columns (total 33 columns):
 #   Column                 Non-Null Count  Dtype
---  ------                 --------------  -----
 0   owner_id               14544 non-null  int64
 1   name                   14542 non-null  object
 2   revised_state          14542 non-null  object
 3   cluster                14542 non-null  object
 4   territory              14542 non-null  object
 5   status                 14542 non-null  object
 6   category               13494 non-null  object
 7   category_repo          13494 non-null  object
 8   Product_group          14544 non-null  object
 9   Revenue                13494 non-null  float64
 10  Sale_Qty               13494 non-null  float64
 11  return_amount          13494 non-null  float64
 12  Ret_Qty                13494 non-null  float64
 13  Net_Rev                13494 non-null  float64
 14  Net_Qty                13494 non-null  float64
 15  ABS_Given_or_not_FY26  14544 non-null  object
 16  ABS_Amt_Fy26           591 non-null    float64
 17  ABS_Given_or_not_FY27  14544 non-null  object
 18  ABS_Amt_Fy27           1616 non-null   float64
 19  TM                     14542 non-null  object
 20  SM                     14542 non-null  object
 21  Rank                   13494 non-null  float64
 22  FY26 Packet            0 non-null      float64
 23  FY27 Packet            0 non-null      float64
 24  Unnamed: 24            0 non-null      float64
 25  Unnamed: 25            0 non-null      float64
 26  Unnamed: 26            0 non-null      float64
 27  Product_group.1        37 non-null     object
 28  Unnamed: 28            12 non-null     float64
 29  Unnamed: 29            12 non-null     float64
 30  Unnamed: 30            12 non-null     float64
 31  Unnamed: 31            0 non-null      float64
 32  Unnamed: 32            52 non-null     object
dtypes: float64(18), int64(1), object(14)
memory usage: 3.7+ MB
```

Cleaning data - Removing Unnamed Columns & Filling missing Values

```
df = df.drop(['Unnamed: 24' , 'Unnamed: 25','Unnamed: 26','Product_group.1','Unnamed: 28','Unnamed: 29','Unnamed: 30' , 'Ur
```

```
df = df[df['revised_state'].notna()]
```

```
df['category'] = df['category'].fillna("Seeds")

df.loc[(df['Product_group'] == 'Crop Care') & (df['category_repo'].isna()), 'category_repo'] = 'CP_CN'

df.loc[(df['Product_group'] != 'Crop Care') & (df['category_repo'].isna()), 'category_repo'] = 'Seeds - FC'

df['Revenue'] = df['Revenue'].fillna( round(df['Revenue'].mean(),0))

cols = ['Sale_Qty', 'return_amount', 'Ret_Qty', 'Net_Rev', 'Net_Qty']

for i in cols:
  df[cols] = df[cols].fillna(round(df[cols].mean(),0))
```

```
df['ABS_Amt_Fy26'] = df['ABS_Amt_Fy26'].fillna(0)

df['ABS_Amt_Fy27'] = df['ABS_Amt_Fy27'].fillna(0)

df = df.round(0)
```

Adding Addional metric Column

```
df['return_pct'] = df['return_amount'] / df['Revenue']
df['net_efficiency'] = df['Net_Rev'] / df['Revenue']
df['abs_flag_fy27'] = df['ABS_Given_or_not_FY27'] == 'YES'
df['abs_flag_fy26'] = df['ABS_Given_or_not_FY26'] == 'YES'
df['Net_Rev_in_lacs'] =  df['Net_Rev']/10**5

df = df.round(0)
```

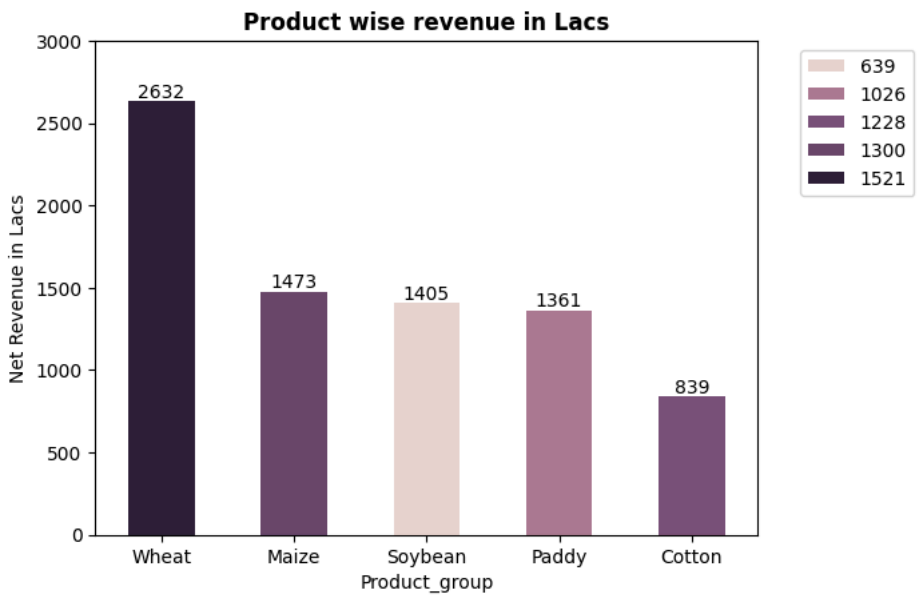
```
# KPI Cards

total_revenue = df['Revenue'].sum()
net_revenue = df['Net_Rev'].sum()
total_qty = df['Sale_Qty'].sum()
unique_partner = df['owner_id'].nunique()
return_pct = df['return_amount'].sum() / df['Revenue'].sum()
abs_adoption = df.loc[df['ABS_Given_or_not_FY27'] == 'YES' ,'ABS_Amt_Fy27' ].mean()
```

Product wise Revenue

Understand the product performence

```
top_product  = df.groupby('Product_group' , as_index= False).agg(  Unique_partner = ('owner_id' , 'nunique' ) ,  Gross_rev

top_product['Gross_rev_in_lac'] = top_product['Gross_rev']/10**5

top_product['Net_rev_in_lac'] = top_product['Net_rev']/10**5

top_product = top_product.round(0)
```

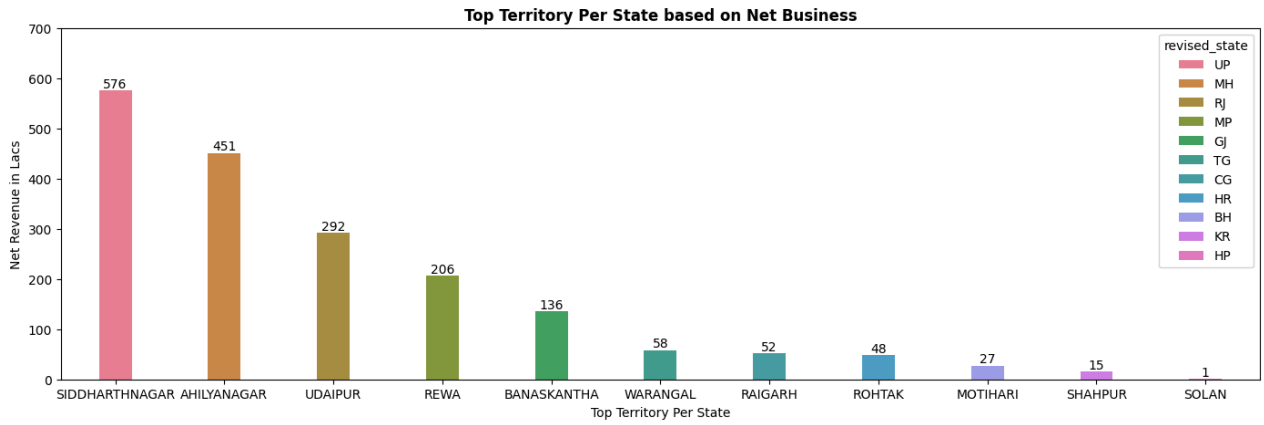
```
ax =sns.barplot(  data = top_product , x = 'Product_group' , y = 'Net_rev_in_lac' , hue = 'Unique_partner' ,  width = 0.5 )
plt.title('Product wise revenue in Lacs' , fontweight = 'bold')
plt.legend(bbox_to_anchor = (1.05 , 1) , loc = 'upper left' )
for i in ax.containers:
  ax.bar_label( i , fontsize = 10)
plt.ylim(0 , 3000)
plt.ylabel("Net Revenue in Lacs")
plt.show()
```



Top Territory Per State by Net Revenue

```
terr__wise = df.groupby( ['revised_state' , 'territory'] , as_index= False)['Net_Rev'].sum()

terr__wise['rank'] = terr__wise.groupby('revised_state' )['Net_Rev'].rank( method = 'dense' , ascending = False ).sort_valu

terr__wise['Net_Rev_in_lac'] = terr__wise['Net_Rev']/10**5

terr__wise = terr__wise[terr__wise['rank'] ==1 ].sort_values(by = 'Net_Rev_in_lac' , ascending= False)

terr__wise = terr__wise.round(0)
```

```
plt.figure(figsize= ( 17 , 5))
ax = sns.barplot( data = terr__wise , x = 'territory' , y = 'Net_Rev_in_lac' , hue = 'revised_state'  , width = 0.3 )

for i in ax.containers:
  ax.bar_label( i , fontsize = 10)
plt.ylim ( 0 , 700)
plt.title("Top Territory Per State based on Net Business" , fontweight = 'bold')
plt.xlabel( 'Top Territory Per State')
plt.ylabel( 'Net Revenue in Lacs')
plt.show()
```

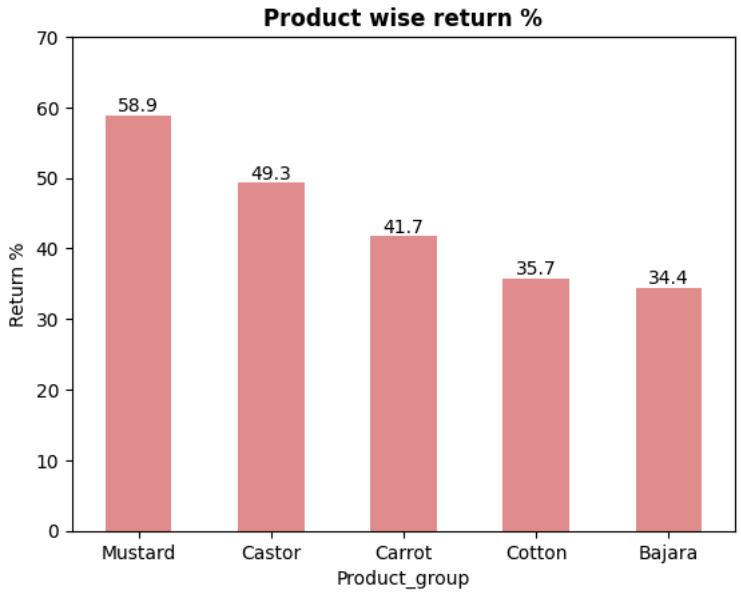


Start coding or generate with AI.

Product Wise Return

```
ret = df.groupby('Product_group' , as_index= False)['return_pct'].mean().sort_values( by = 'return_pct' , ascending= False

ret['return_pct'] = round(ret['return_pct']*100,1)

ret = ret.head()
```

```
ax =sns.barplot(  data = ret , x = 'Product_group' , y = 'return_pct'  ,  width = 0.5  , color = 'lightcoral')
plt.title('Product wise return %' , fontweight = 'bold')
for i in ax.containers:
  ax.bar_label( i , fontsize = 10)
plt.ylim(0 , 70)
plt.ylabel("Return %")
plt.show()
```

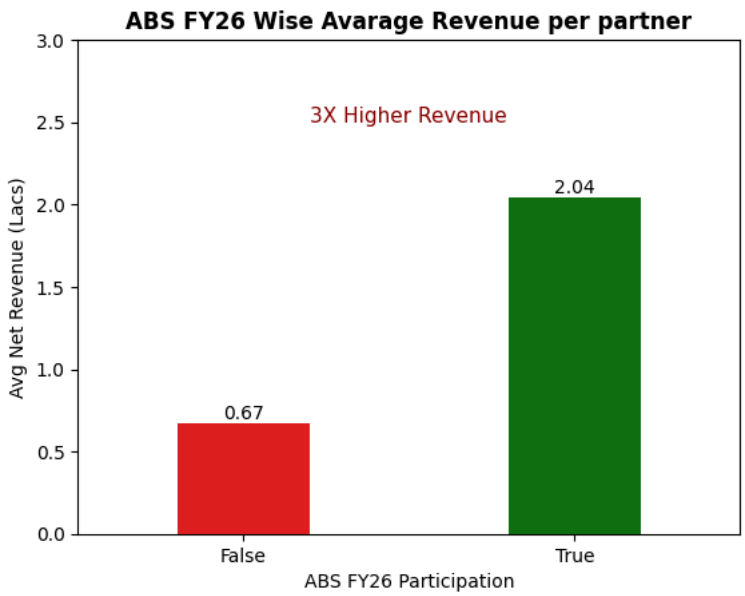



```
df.columns
```

```
Index(['owner_id', 'name', 'revised_state', 'cluster', 'territory', 'status',
       'category', 'category_repo', 'Product_group', 'Revenue', 'Sale_Qty',
       'return_amount', 'Ret_Qty', 'Net_Rev', 'Net_Qty',
       'ABS_Given_or_not_FY26', 'ABS_Amt_Fy26', 'ABS_Given_or_not_FY27',
       'ABS_Amt_Fy27', 'TM', 'SM', 'return_pct', 'net_efficiency',
       'abs_flag_fy27', 'abs_flag_fy26', 'Net_Rev_in_lacs'],
      dtype='object')
```

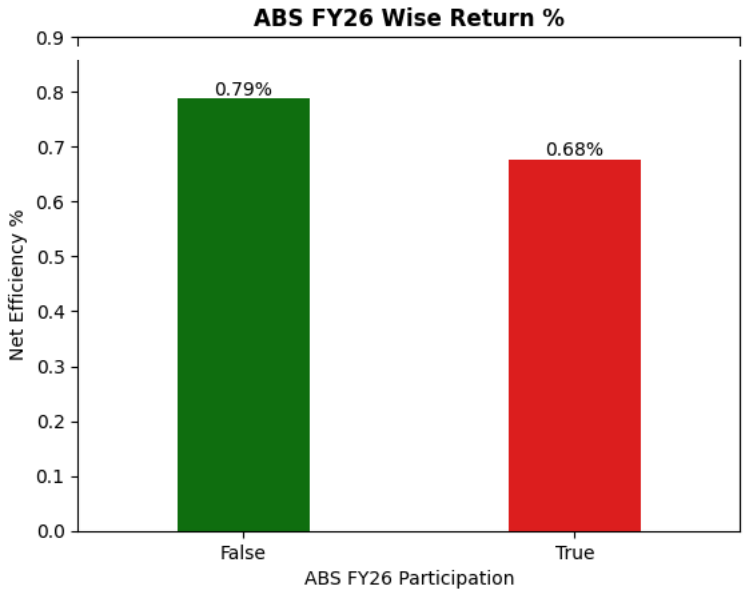
ABS VS NON ABS Partner Performence in FY26 (Benchmark for FY27)

```
ax=sns.barplot( data = df , x = 'abs_flag_fy26' , y = 'Net_Rev_in_lacs' , width= 0.4 , errorbar= None , palette=['red', 'gr

for i in ax.containers:
  ax.bar_label( i , fontsize = 10  , fmt= '%.2f' )

plt.title( 'ABS FY26 Wise Avarage Revenue per partner' , fontweight = 'bold')
plt.ylim( 0 , 3)
plt.text(0.5, 2.5, '3X Higher Revenue', ha='center', fontsize=11, color='darkred')
palette = ['#f4a6a6', '#b30000']
plt.ylabel('Avg Net Revenue (Lacs)')
plt.xlabel('ABS FY26 Participation')
plt.show()
```




```
ax=sns.barplot( data = df , x = 'abs_flag_fy26' , y = 'net_efficiency' , width= 0.4 , errorbar= None ,  palette=['green',
```

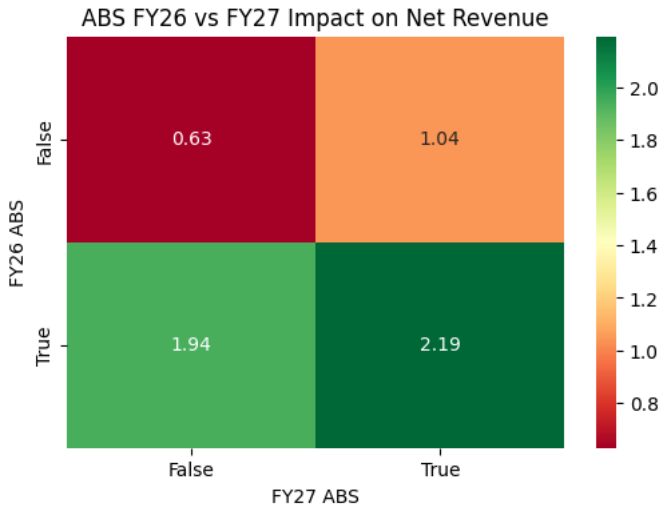
```
for i in ax.containers:
  ax.bar_label( i , fontsize = 10  , fmt= '%0.2f%%' )

plt.title( 'ABS FY26 Wise Return %' , fontweight = 'bold')

plt.ylim( 0 , 0.9)

plt.xlabel('ABS FY26 Participation')
plt.ylabel( 'Net Efficiency %')


plt.show()
```



ABS FY26 VS FY27

```
abs_grp = df.groupby(['abs_flag_fy26', 'abs_flag_fy27'])['Net_Rev_in_lacs'] \
    .mean() \
    .reset_index()

heatmap_data = abs_grp.pivot(
    index='abs_flag_fy26',
    columns='abs_flag_fy27',
    values='Net_Rev_in_lacs'
)
```

```
plt.figure(figsize=(6,4))

sns.heatmap(
    heatmap_data,
    annot=True,
    fmt=".2f",
    cmap="RdYlGn"  )

plt.title('ABS FY26 vs FY27 Impact on Net Revenue')
plt.xlabel('FY27 ABS')
plt.ylabel('FY26 ABS')

plt.show()
```




```
from google.colab import files

download =  df.to_csv("ABS_RAW.csv" , index= None)

files.download("ABS_RAW.csv")
```